Name : Mayank Heda
Roll No. 2401035
T12 - AI&DS
Subject : ASD&D Lab

EXPERIMENT – 5

AIM : To automate application deployment using Github/Jenkins.

LO MAPPED : LO3.

LO3 : Build and manage automated CI/CD pipelines for application development, testing,

Packaging and deployment.

THEORY :

Continuous Integration and Continuous Deployment

CI/CD is a software development methodology that automates the process of taking source-code changes from development to a deployed application. It minimizes manual tasks and makes software delivery faster and more reliable.

Continuous Integration (CI):

Continuous Integration involves regularly adding developers' code changes to a common repository. Each change can be automatically built and tested, which helps detect errors early and keeps the project in a stable state.

Continuous Deployment (CD):

Continuous Deployment is the next stage of the CI process, where code that successfully passes the required checks is automatically delivered to the target environment. This avoids the need for manually transferring application files.

Jenkins:

Jenkins is an open-source automation tool used for performing tasks such as building, testing, and deploying applications. It can connect with source-code repositories like GitHub and execute the stages defined in a pipeline.

Jenkins Declarative Pipeline:

A Declarative Pipeline defines the CI/CD process in a structured manner using a Jenkinsfile. Different stages such as Build, Test, and Deploy can be specified in the file. Storing this configuration along with the application source code makes the process repeatable.

GitHub:

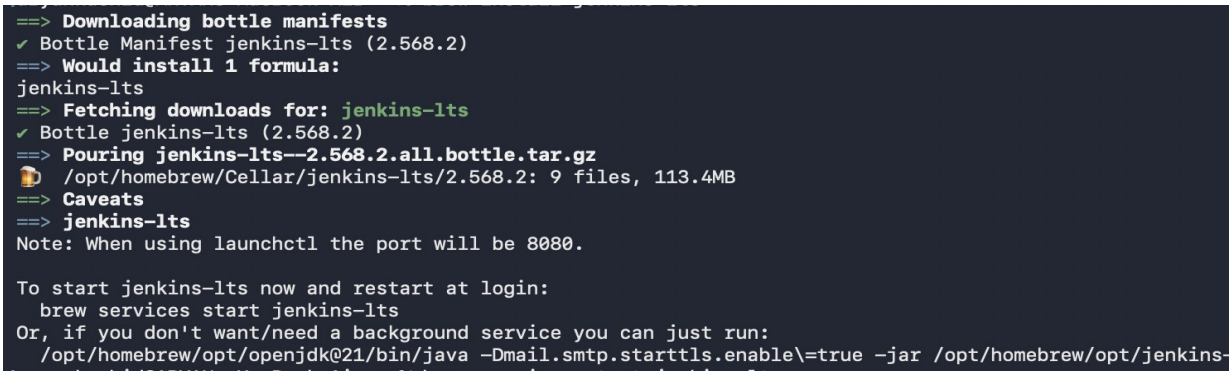
GitHub is a remote repository used to store and manage application source code. In this experiment, Jenkins obtains the application files and pipeline configuration from the GitHub repository.

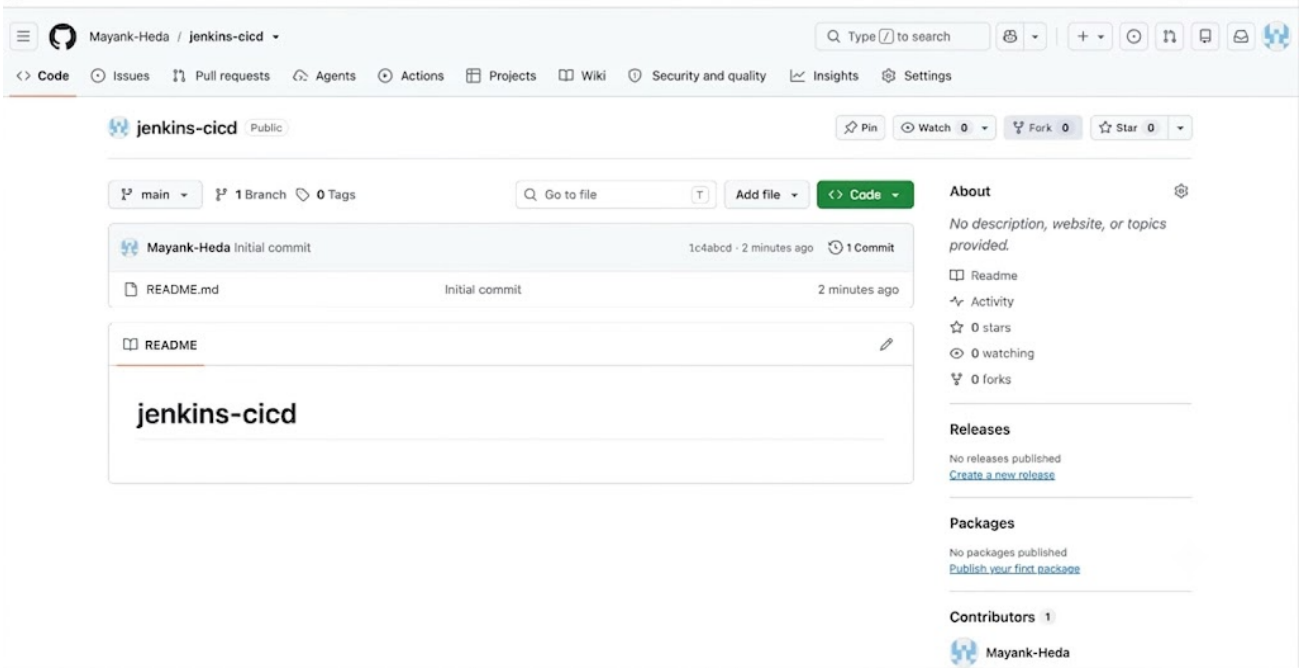
CI/CD Flow:

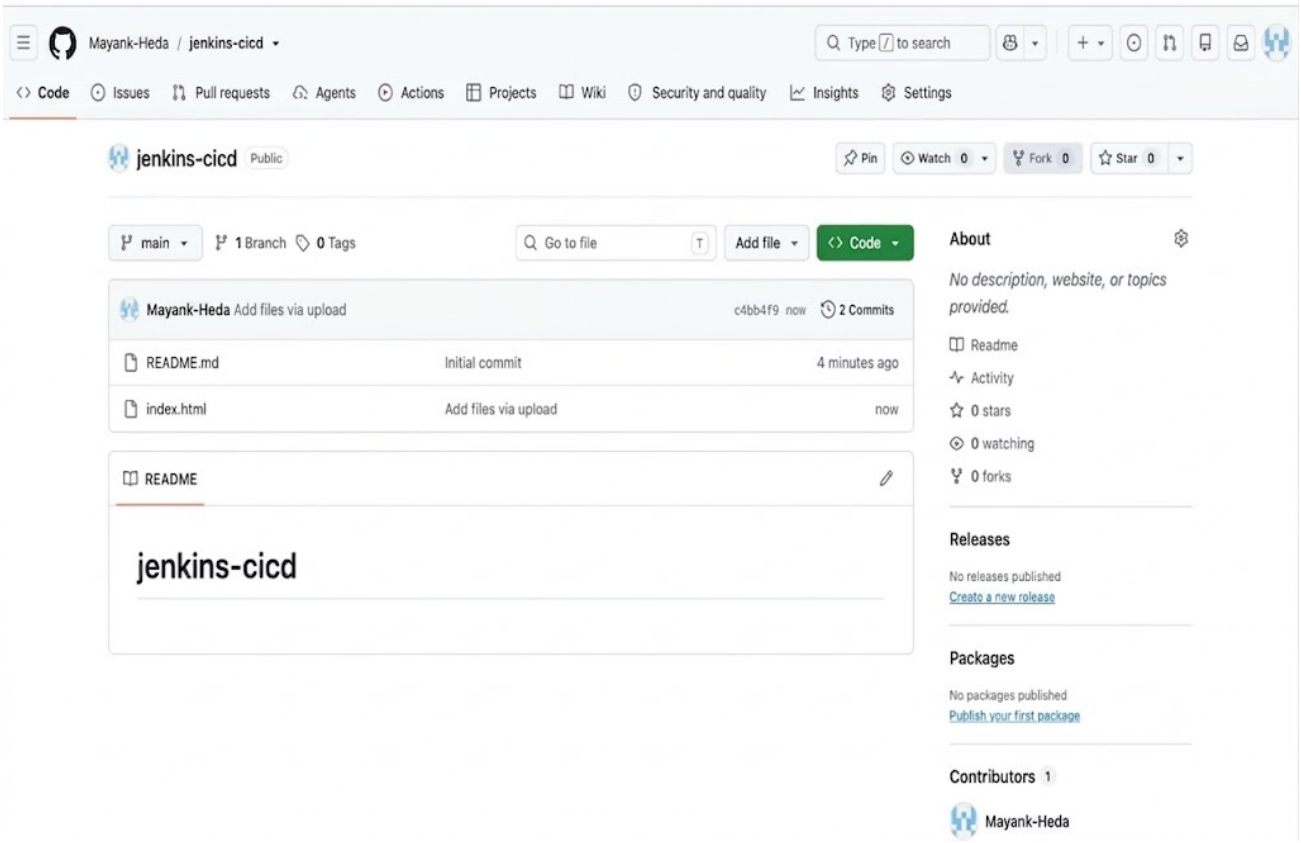
Developer modifies code → GitHub Repository → Jenkins fetches code → Build → Test → Deploy → Updated Application

EXECUTION :

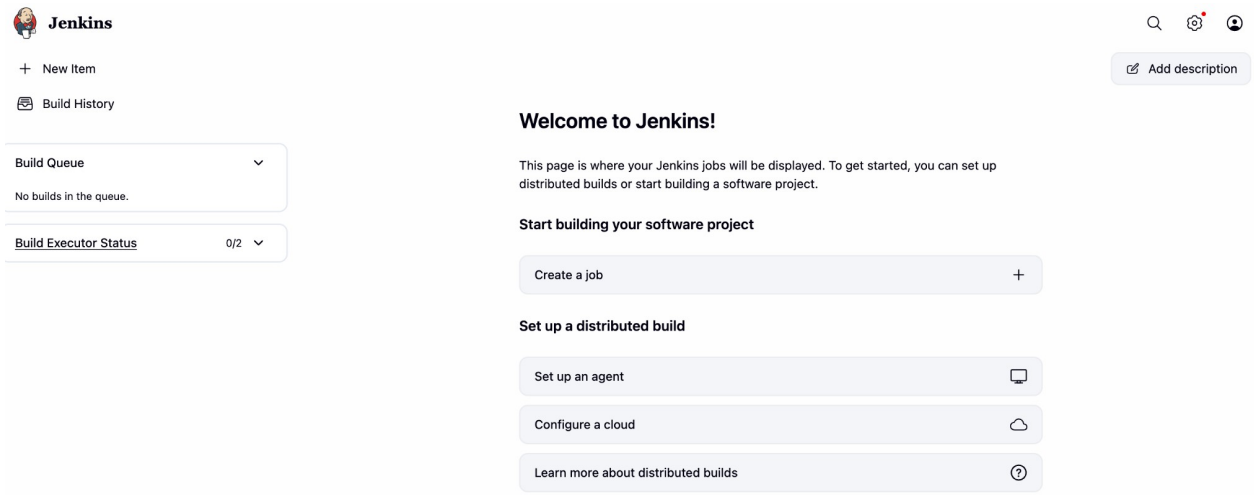
Step 1: Jenkins Initialization and Setup

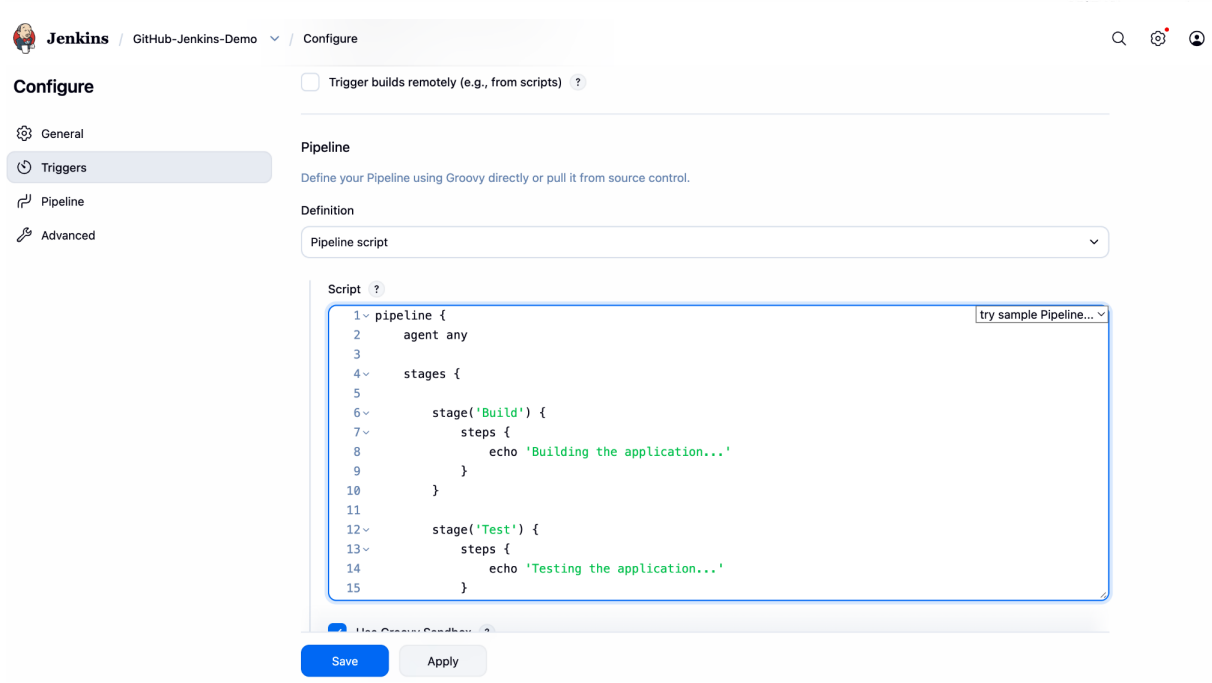



Mayank-Heda / jenkins-cicd
Type / to search
Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
jenkins-cicd
Public
Pin
Watch 0
Fork 0
Star 0
main
1 Branch
0 Tags
Go to file
Add file
Code
Mayank-Heda Initial commit
1c4abcd · 2 minutes ago
1 Commit
README.md
Initial commit
2 minutes ago
README
jenkins-cicd
About
No description, website, or topics provided.
Readme
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
Contributors 1
Mayank-Heda

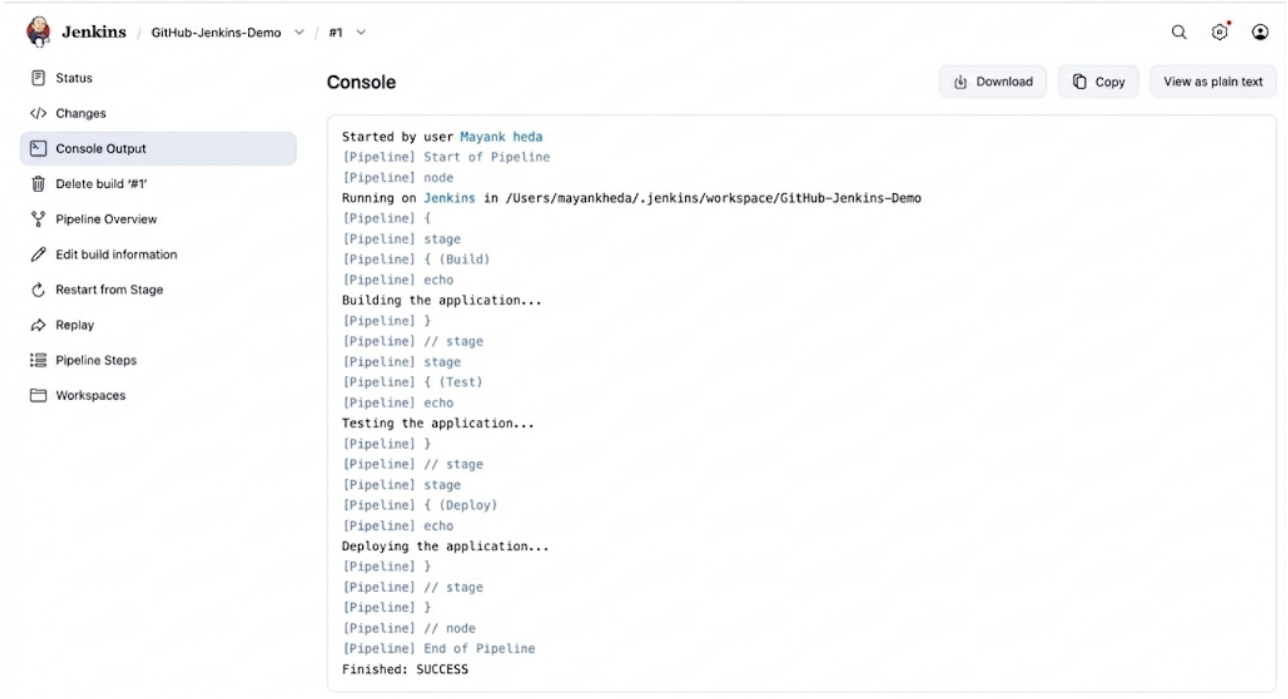

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Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
jenkins-cicd
Public
Pin
Watch 0
Fork 0
Star 0
main
1 Branch
0 Tags
Go to file
Add file
Code
Mayank-Heda Add files via upload
c4bb4f9 now
2 Commits
README.md
Initial commit
4 minutes ago
index.html
Add files via upload
now
README
jenkins-cicd
About
No description, website, or topics provided.
Readme
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
Contributors 1
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Step 2: Jenkins Pipeline Creation and Initial Validation

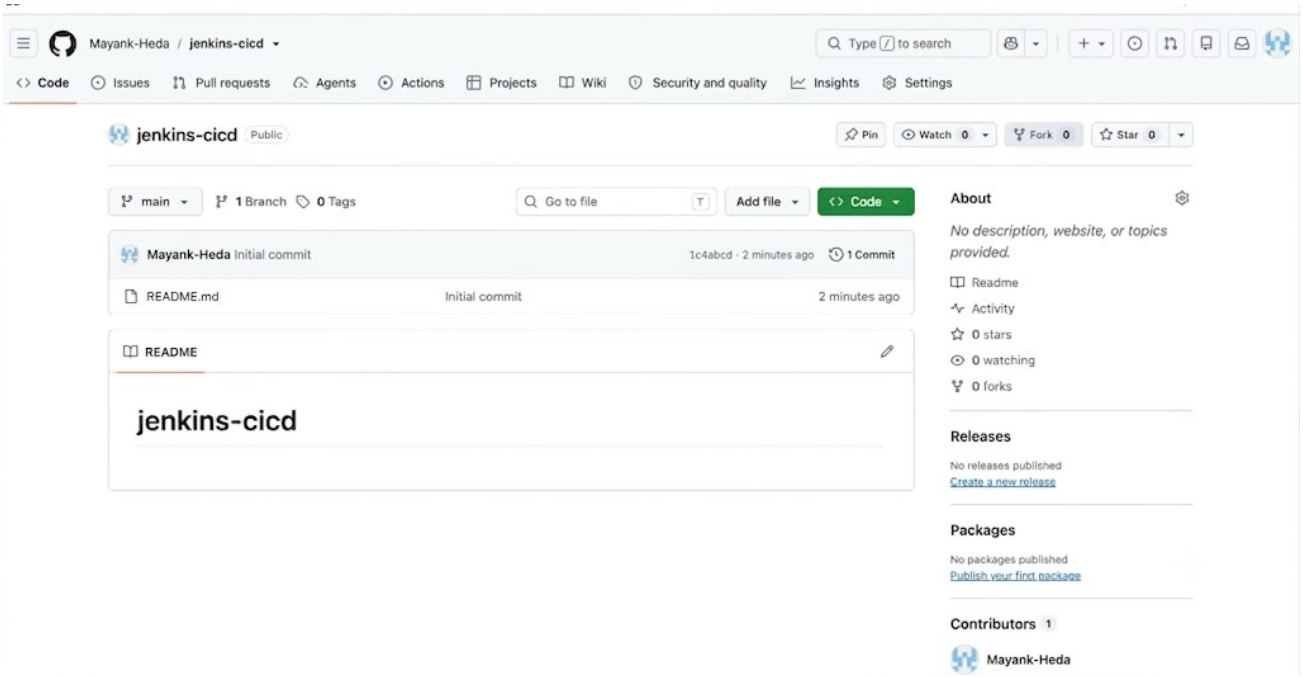


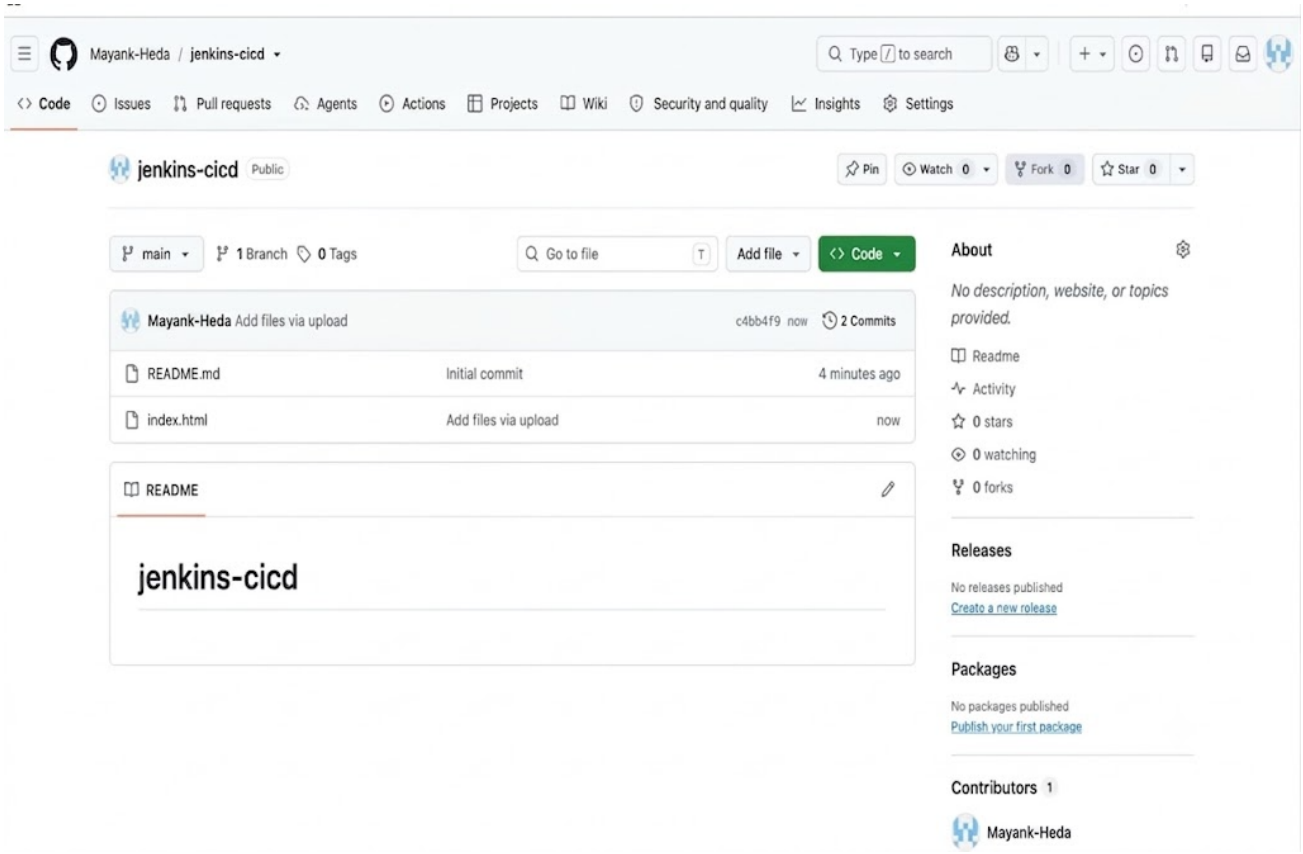


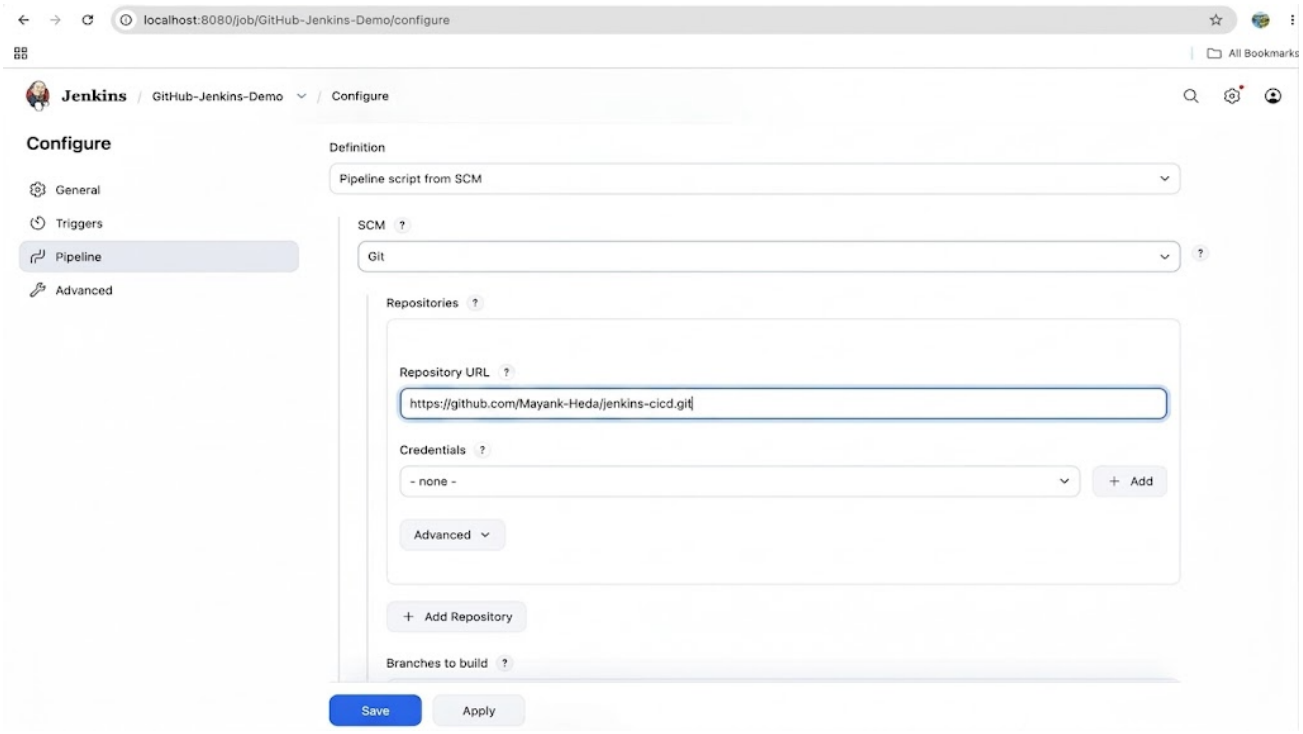
Step 3: Initial Pipeline Execution

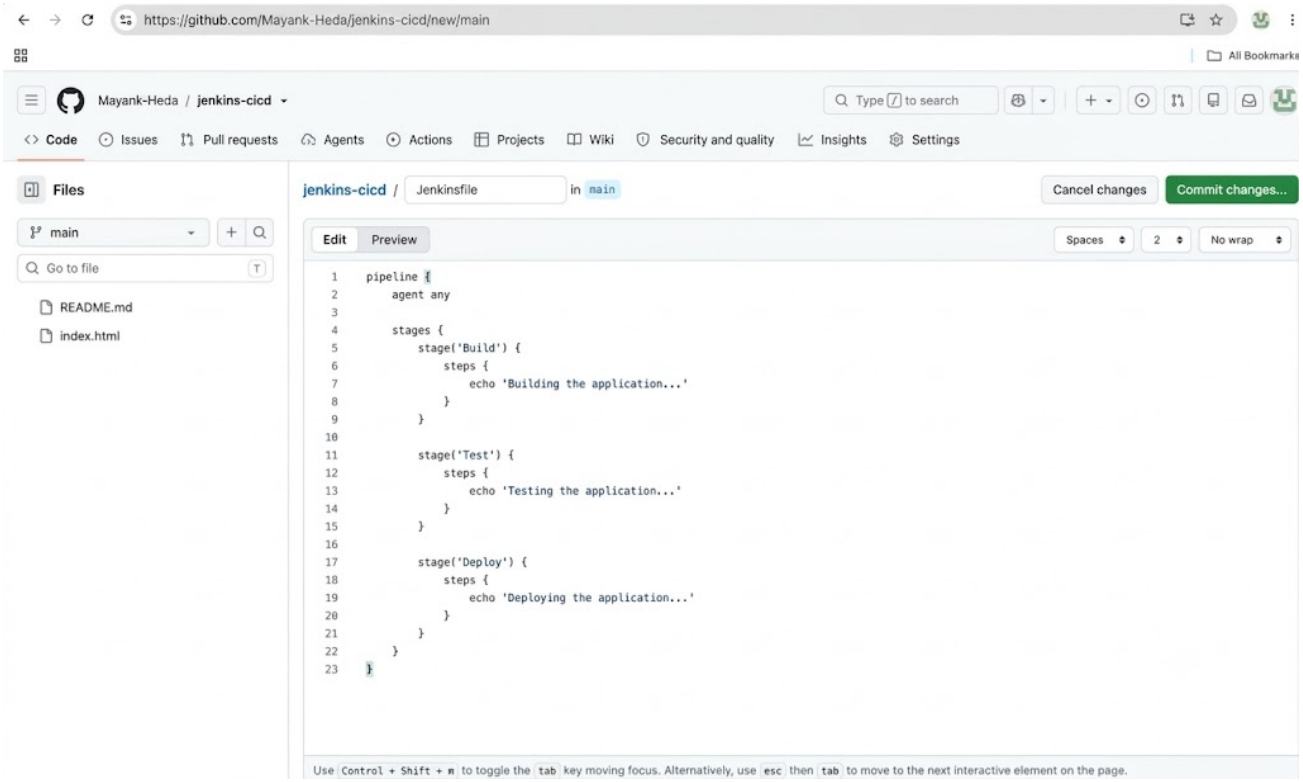


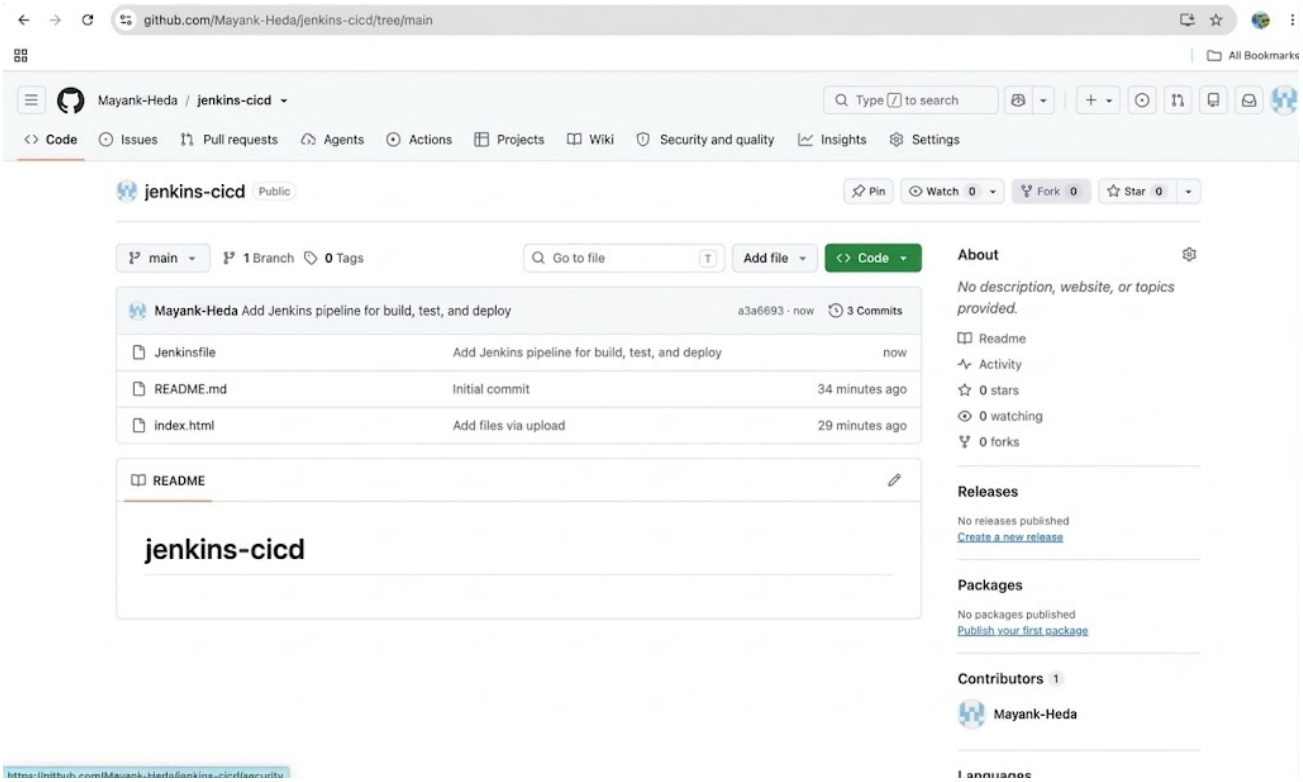
Step 4: GitHub Repository and Jenkinsfile Configuration




Mayank-Heda / jenkins-cicd
Type / to search
Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
jenkins-cicd
Public
Pin
Watch 0
Fork 0
Star 0
main
1 Branch
0 Tags
Go to file
T
Add file
Code
Mayank-Heda Add files via upload
c4bb4f9 now
2 Commits
README.md
Initial commit
4 minutes ago
index.html
Add files via upload
now
README
jenkins-cicd
About
No description, website, or topics provided.
Readme
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
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localhost:8080/job/GitHub-Jenkins-Demo/configure
All Bookmarks
Jenkins
GitHub-Jenkins-Demo
Configure
Configure
General
Triggers
Pipeline
Advanced
Definition
Pipeline script from SCM
SCM
Git
Repositories
Repository URL
https://github.com/Mayank-Heda/jenkins-cicd.git
Credentials
- none -
Add
Advanced
Add Repository
Branches to build
Save
Apply


https://github.com/Mayank-Heda/jenkins-cicd/new/main
Mayank-Heda / jenkins-cicd
Type / to search
Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
Files
main
Go to file
README.md
index.html
jenkins-cicd / Jenkinsfile in main
Cancel changes
Commit changes...
Edit
Preview
Spaces
2
No wrap
pipeline {
agent any
stages {
stage('Build') {
steps {
echo 'Building the application...'
}
}
stage('Test') {
steps {
echo 'Testing the application...'
}
}
stage('Deploy') {
steps {
echo 'Deploying the application...'
}
}
}
}
Use Control + Shift + m to toggle the tab key moving focus. Alternatively, use esc then tab to move to the next interactive element on the page.


github.com/Mayank-Heda/jenkins-cicd/tree/main
Mayank-Heda / jenkins-cicd
Type / to search
Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
jenkins-cicd Public
Pin
Watch 0
Fork 0
Star 0
main
1 Branch
0 Tags
Go to file
Add file
Code
Mayank-Heda Add Jenkins pipeline for build, test, and deploy
a3a6693 · now
3 Commits
Jenkinsfile
Add Jenkins pipeline for build, test, and deploy
now
README.md
Initial commit
34 minutes ago
index.html
Add files via upload
29 minutes ago
README
jenkins-cicd
About
No description, website, or topics provided.
Readme
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
Contributors 1
Mayank-Heda
Languages

Step 5: GitHub Checkout and Automated Deployment



Files

main

Go to file T

Jenkinsfile

README.md

index.html

jenkins-cicd / Jenkinsfile

Mayank-Heda Enhance deployment stage with file operations

a057dfe · now History

Code Blame 28 lines (24 loc) · 589 Bytes Raw

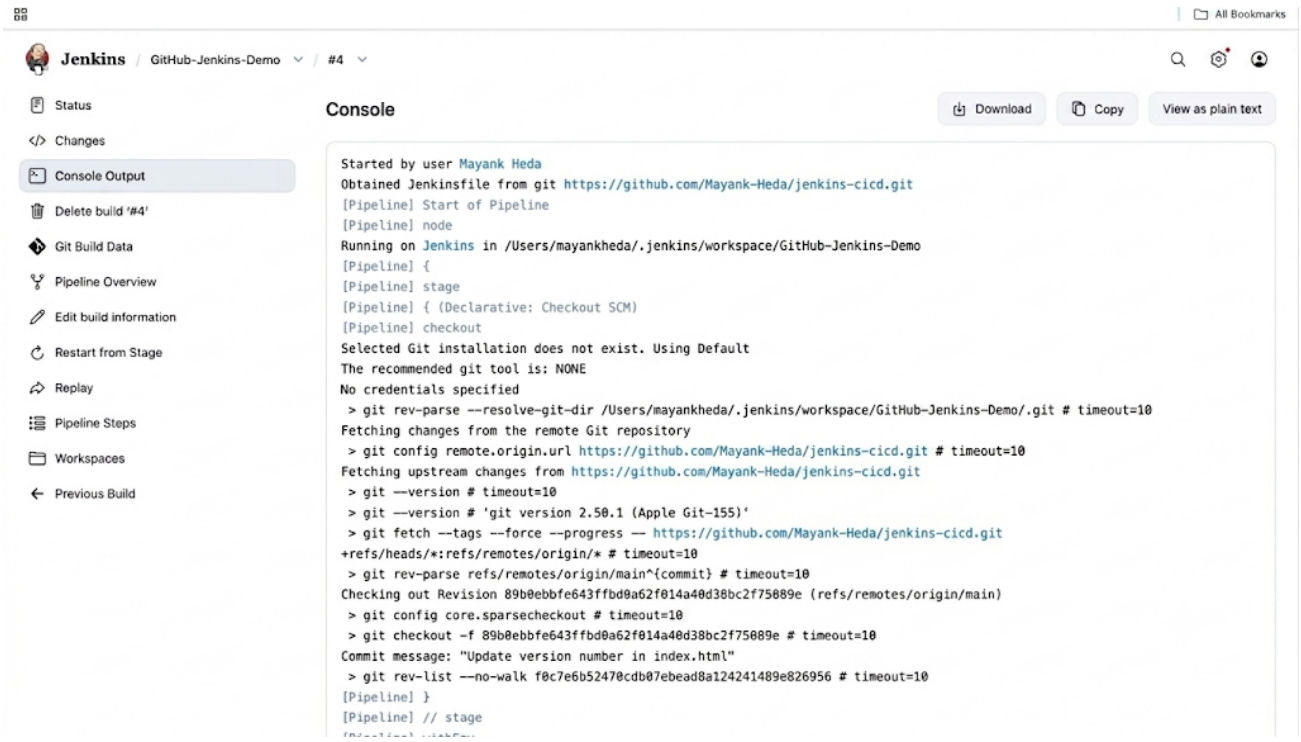
```
pipeline {
    agent any

    stages {

        stage('Build') {
            steps {
                echo 'Building the application...'
            }
        }

        stage('Test') {
            steps {
                echo 'Testing the application...'
            }
        }

        stage('Deploy') {
            steps {
                sh '''
                    mkdir -p "$HOME/Deployments/DemoApp"
                    cp index.html "$HOME/Deployments/DemoApp/index.html"
                '''
                echo 'Application deployed successfully.'
            }
        }
    }
}
```



Jenkins / GitHub-Jenkins-Demo / #4

Status

Changes

Console Output

Delete build '#4'

Git Build Data

Pipeline Overview

Edit build information

Restart from Stage

Replay

Pipeline Steps

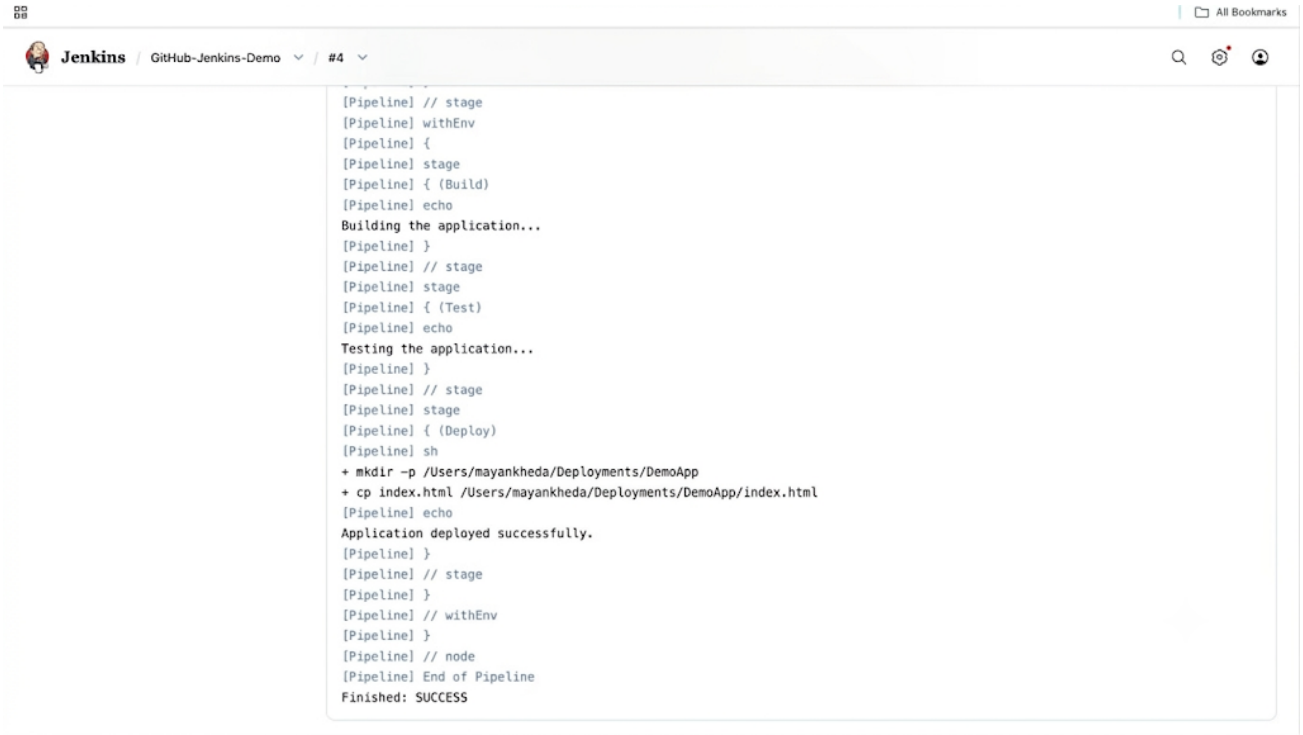
Workspaces

Previous Build

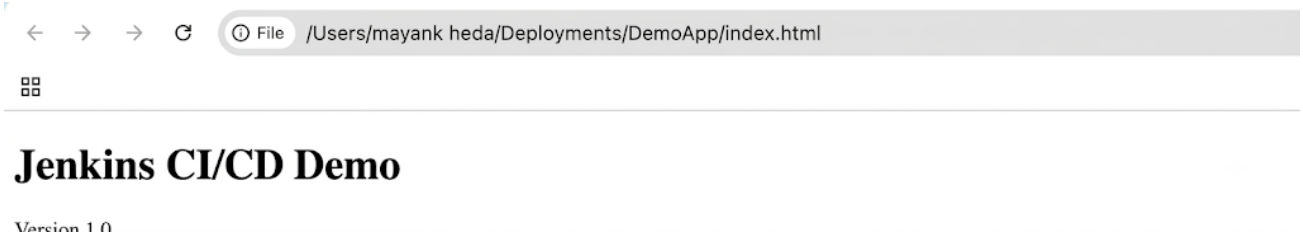
Console

Download Copy View as plain text

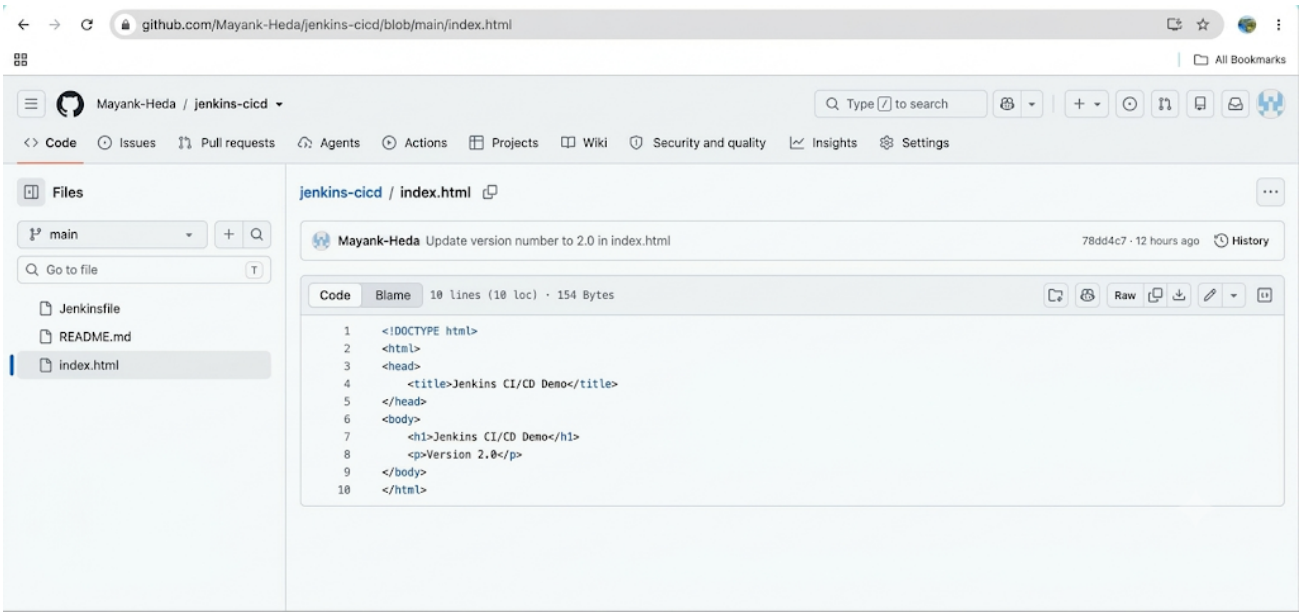
```
Started by user Mayank Heda
Obtained Jenkinsfile from git https://github.com/Mayank-Heda/jenkins-cicd.git
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /Users/mayankheda/.jenkins/workspace/GitHub-Jenkins-Demo
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
No credentials specified
 > git rev-parse --resolve-git-dir /Users/mayankheda/.jenkins/workspace/GitHub-Jenkins-Demo/.git # timeout=10
Fetching changes from the remote Git repository
 > git config remote.origin.url https://github.com/Mayank-Heda/jenkins-cicd.git # timeout=10
Fetching upstream changes from https://github.com/Mayank-Heda/jenkins-cicd.git
 > git --version # timeout=10
 > git --version # 'git version 2.50.1 (Apple Git-155)'
 > git fetch --tags --force --progress -- https://github.com/Mayank-Heda/jenkins-cicd.git
+refs/heads/*:refs/remotes/origin/* # timeout=10
 > git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 89b0ebbfe643ffbd0a62f014a40d38bc2f75089e (refs/remotes/origin/main)
 > git config core.sparsecheckout # timeout=10
 > git checkout -f 89b0ebbfe643ffbd0a62f014a40d38bc2f75089e # timeout=10
Commit message: "Update version number in index.html"
 > git rev-list --no-walk f0c7e6b52470cdb07ebead8a124241489e826956 # timeout=10
[Pipeline] }
[Pipeline] // stage
```

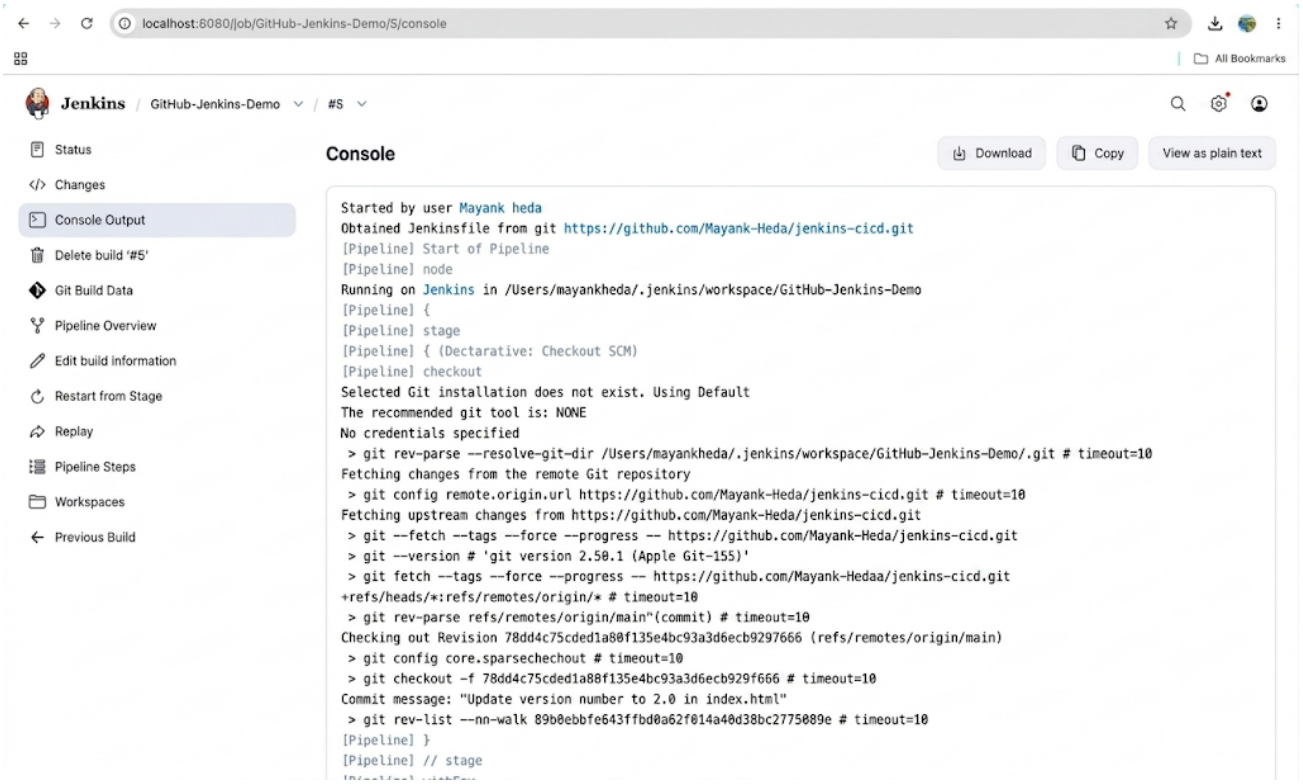


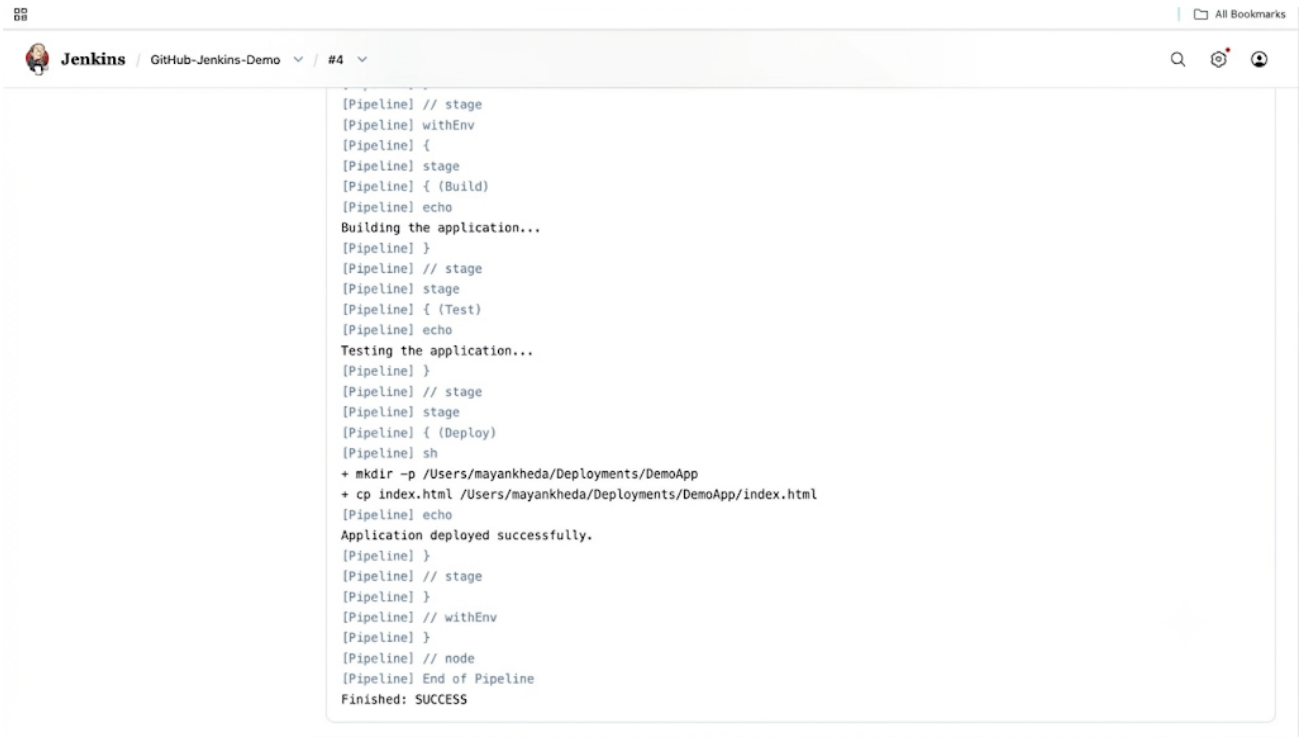
Step 6: Initial Application Deployment



Step 7: CI/CD Update and Redeployment


github.com/Mayank-Heda/jenkins-cicd/blob/main/index.html
Mayank-Heda / jenkins-cicd
Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
Files
main
Go to file
Jenkinsfile
README.md
index.html
jenkins-cicd / index.html
Mayank-Heda Update version number to 2.0 in index.html
78dd4c7 · 12 hours ago
History
Code
Blame
10 lines (10 loc) · 154 Bytes
Raw
<!DOCTYPE html>
<html>
<head>
<title>Jenkins CI/CD Demo</title>
</head>
<body>
<h1>Jenkins CI/CD Demo</h1>
Version 2.0
</body>
</html>


localhost:8080/job/GitHub-Jenkins-Demo/5/console
Jenkins
GitHub-Jenkins-Demo
#5
Status
Changes
Console Output
Delete build '#5'
Git Build Data
Pipeline Overview
Edit build information
Restart from Stage
Replay
Pipeline Steps
Workspaces
Previous Build
Console
Download
Copy
View as plain text
Started by user Mayank heda
Obtained Jenkinsfile from git https://github.com/Mayank-Heda/jenkins-cicd.git
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /Users/mayankheda/.jenkins/workspace/GitHub-Jenkins-Demo
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Dectarative: Checkout SCM)
[Pipeline] checkout
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
No credentials specified
> git rev-parse --resolve-git-dir /Users/mayankheda/.jenkins/workspace/GitHub-Jenkins-Demo/.git # timeout=10
Fetching changes from the remote Git repository
> git config remote.origin.url https://github.com/Mayank-Heda/jenkins-cicd.git # timeout=10
Fetching upstream changes from https://github.com/Mayank-Heda/jenkins-cicd.git
> git --fetch --tags --force --progress -- https://github.com/Mayank-Heda/jenkins-cicd.git
> git --version # 'git version 2.50.1 (Apple Git-155)'
> git fetch --tags --force --progress -- https://github.com/Mayank-Hedaa/jenkins-cicd.git
+refs/heads/*:refs/remotes/origin/* # timeout=10
> git rev-parse refs/remotes/origin/main^(commit) # timeout=10
Checking out Revision 78dd4c75cded1a88f135e4bc93a3d6ecb9297666 (refs/remotes/origin/main)
> git config core.sparsechechout # timeout=10
> git checkout -f 78dd4c75cded1a88f135e4bc93a3d6ecb929f666 # timeout=10
Commit message: "Update version number to 2.0 in index.html"
> git rev-list --nn-walk 89b0ebbfe643ffbd0a62f014a40d38bc2775089e # timeout=10
[Pipeline] }
[Pipeline] // stage



Step 8: Final Deployment Verification



CONCLUSION :

The experiment successfully demonstrated how GitHub and Jenkins can be combined to automate application deployment. By using a Declarative Jenkinsfile, the Build, Test, and Deploy tasks were organized into a repeatable pipeline, reducing the need for manual deployment. The initial application as well as its modified version were successfully deployed through the pipeline. This showed that changes made in the GitHub repository could be automatically processed by Jenkins and delivered to the deployment environment. Overall, the experiment provided practical knowledge of CI/CD, Pipeline as Code, GitHub integration, and automated software deployment.